

Start of Life

Eagerness

How to evaluate an async function application.

LAZY

Evaluate to a coroutine without executing further.

Python, Rust

EAGER

Evaluate in current thread, and schedule as task on await.

C#, JavaScript

SEMI-EAGER

Evaluate to task, and schedule task for execution.

Swift

Suspension

Guarantees on whether await points suspend.

STATIC

Await points guaranteed to suspend.

JavaScript

DYNAMIC

No guarantees on awaiting tasks.

Python, Rust, C#, Swift

End of Life

Extent

The default interval of time during which a task may exist.

INDEFINITE

Tasks by default may exist until the end of the runtime.

JavaScript, C#, Tokio, Smol, Asyncio

DYNAMIC

Tasks by default may exist until the end of their spawning scope.

Swift, Trio

Reference Strength

[For indefinite extent]
The type of reference to a task held by the runtime.

STRONG

The runtime holds a strong reference.

JavaScript, C#, Tokio

WEAK

The runtime holds a weak reference.

Asyncio, Smol

Destruction

How a task is cleaned up at the end of its extent.

AWAITED

The task is awaited to completion.

JavaScript, Trio

CANCELLED

The task is cancelled, and then possibly awaited.

Swift, Tokio, Smol, Asyncio

TERMINATED

The program exits.

C#

Propagation

What happens to exceptions in unawaited tasks.

DESTRUCTION

Exceptions are reraised by the task parent.

Trio

NEVER

The exception is kept within the task.

JavaScript, C#, Tokio, Smol, Asyncio, Swift

Cancellation

Awareness

Whether a task is able to respond to being cancelled.

UNAWARE

The task cannot respond to being cancelled.

Rust

AWARE

The task can respond to being cancelled.

Asyncio, Trio, Swift

Direction

How cancellation is communicated through the task graph.

TOP-DOWN

Starting from the root task, and down through the children.

Rust

BOTTOM-UP

Starting from the leaf tasks, and up through the parents.

Asyncio, Trio

SIMULTANEOUS

To all nodes in the task graph at once.

Swift

Persistence

[For aware cancellation]
How long a cancellation of a task lasts.

TRANSIENT

A task can ignore cancellation and proceed as normal.

Asyncio

PERSISTENT

A task can ignore cancellation but remains cancelled.

Trio, Swift